

Counting Stable Matchings: One-Sided Entropy and Sharp Stable-Pair Bounds

Huibo Xu
Nanyang Technological University

Abstract

Stable matching pairs participants on two sides so that no pair would prefer each other to their assigned partners. Although one stable outcome can be found efficiently, determining how many can coexist under the same preferences is a long-standing question of Knuth. Let $\text{SM}(n)$ be the maximum number of stable matchings over all strict, complete instances with n men and n women. We prove $\text{SM}(n) < 3.3248^n$ for every $n \geq 1$, improving the previous upper bound $3.55^n + O(1)$. We develop a one-sided entropy encoding in which a revealed matching edge gives an oriented cut valid across all stable completions of the revealed prefix. These cuts enclose each conditional partner set in an interval whose width, given the target's random reveal priority, is dominated by the sum of two independent geometric waiting times minus one. This reduction bounds the entropy of each of the n male partner coordinates. Local endpoint and rotation witnesses give a fixed expected loss at each target where they occur. A deletion argument on rotation boundaries shows that a uniformly random stable matching has linearly many such targets in expectation, yielding a linear improvement in the total entropy bound.

A second theorem gives the sharp bound $|\text{Stab}(I)| \leq 2^{m/4}$, where m counts distinct pairs appearing in stable matchings. The bound has an additional exponential penalty for rotations of length greater than two, where rotations are the elementary cyclic exchanges between stable matchings. Equality in the uncorrected bound holds precisely when the rotation poset is an antichain of disjoint binary rotations covering the men. The proof converts participant conflicts between rotations into hypergraph matching constraints and establishes an incidence-budget counting bound for labeled multihypergraphs with no isolated double-edge component. Participant chains and monotone partner movement verify this condition for the rotation hypergraph. As a consequence of the participant-count bound, Kavitha's existing Popular Roommates algorithm runs in $O^*(6.9717^N)$ time on N participants.

1 Introduction

Stable matching studies how to match participants on two sides according to their preferences. Its classical settings include college admissions and marriage, where an assignment must account not only for feasibility but also for whether participants would prefer to change partners. We study the one-to-one stable-marriage model: the two sides have equal size, and every participant strictly ranks everyone on the other side. A perfect matching assigns exactly one partner to each participant. It is *stable* if no two participants on opposite sides who are not matched to each other both prefer one another to their assigned partners. Such a pair is called a *blocking pair*; stability rules out a reassignment in which the two participants both obtain a preferred partner. Gale and Shapley proved that, with strict and complete preferences, a stable matching always exists and can be found efficiently [GS62].

This classical result explains how to obtain one stable outcome, but does not describe the space of all stable outcomes. The same preference profile may admit a unique stable matching or many different ones. We study the following extremal question: with n participants on each side, how many stable matchings can a single preference profile admit? Denote the two sides by M and W , traditionally called men and women. Write $\text{Stab}(I)$ for the stable matchings of an instance I , and put $\text{SM}(n) = \max_I |\text{Stab}(I)|$, where the maximum ranges over all strict, complete instances with n participants on each side. This number measures the choice that remains under stability and bounds the solution space explored by algorithms that use stable-matching enumeration as a subroutine. Knuth posed the determination of the growth of $\text{SM}(n)$ in 1976 [Knu76]. There are $n!$ perfect matchings in total, whereas the best known stable-matching constructions give an exponential base about 2.28 [IL86, GI89, Thu02]. Karlin, Oveis Gharan, and Weber proved the first upper bound of the form c^n for an absolute constant c [KGW18]. Their result shows that stability reduces the worst-case solution space from factorial to exponential size. Palmer and Pálvölgyi subsequently improved the bound to $\text{SM}(n) \leq 3.55^n + O(1)$ [PP21]. A substantial gap between the exponential rates remains. Further progress requires a finer understanding of how stability restricts combinations of local choices, beyond estimating the number of available partners for each participant.

Even two men and two women can admit a choice. Suppose u_1 prefers w_1 to w_2 and u_2 prefers w_2 to w_1 , while w_1 prefers u_2 and w_2 prefers u_1 . Both the matching $\{u_1w_1, u_2w_2\}$ and the matching $\{u_1w_2, u_2w_1\}$ are stable. Moving from one to the other improves both women and hurts both men. Larger instances can contain many interacting versions of this switch, but these switches generally cannot be chosen independently: one may become available only after another has been performed. The difficulty is therefore not only to count each participant’s partner choices, but to determine which choices can form a stable matching together. The relation between local choice and global compatibility is central to the counting problem. Our approach is to retain participant constraints that make these choices compatible or incompatible. For our main result, a bound in terms of n , we use fixed matching edges to constrain the entire family of remaining stable completions. These constraints support a one-sided entropy encoding; a global deletion argument then shows that its local bounds cannot be simultaneously near-tight. An independent second proof retains the participant sets of elementary exchanges. It gives a sharp bound in terms of *stable pairs*—pairs occurring in at least one stable matching—together with an exact equality characterization. These results improve the rate in n and determine the optimal uniform rate in the stable-pair count m .

1.1 Our results

Theorem 1.1 (Main result). *For every integer $n \geq 1$,*

$$\text{SM}(n) < \left(\frac{2078}{625}\right)^n = 3.3248^n.$$

The proof first bounds each conditional partner support by a random interval, giving an exponential base $C_* < 3.331974$. Local witnesses and a deletion argument on rotation boundaries then show that these interval bounds incur a linear total entropy loss, reducing the base to 3.3248. Disjoint replication extends the resulting large-instance bound to every n (Section 6.3). The optimal exponential rate remains open.

Our second result uses a different measure of instance size. A *stable pair* is an edge occurring in some stable matching, so its count m measures the total supply of partner choices rather than the number of participants. The two-by-two example above uses four stable pairs for one binary choice. Our second theorem shows that this rate is optimal, and characterizes when the choices combine without loss. To state the refinement, recall that the elementary cyclic exchanges between stable matchings are called *rotations*. They have precedence constraints: some exchanges must occur before others. These constraints form the rotation poset $\mathcal{R}(I)$; an antichain is a set of rotations with no precedence relation between any two of them. A rotation ρ moves $\ell(\rho) \geq 2$ men to their next stable partners. The *excess rotation length* is $\text{ex}(I) = \sum_{\rho \in \mathcal{R}(I)} (\ell(\rho) - 2)$.

Theorem 1.2 (Sharp stable-pair bound). *Every strict complete instance satisfies*

$$|\text{Stab}(I)| \leq 2^{m/4} \beta^{\text{ex}(I)} \leq 2^{m/4}, \quad \beta = \frac{9}{8^{2^{1/4}}} < 1. \quad (1)$$

Equality in the uncorrected bound holds precisely when the rotation poset is an antichain of length-two rotations whose male participant pairs partition M . For every $t \geq 1$, equality is attained with $n = 2t$, $m = 4t$, and 2^t stable matchings.

Thus $2^{1/4}$ is the optimal uniform base for a bound b^m , with a strict exponential gain whenever the excess is linear in m . In ordinary terms, equality requires disjoint two-person exchanges covering the men, with no precedence constraints between exchanges. Having four available pairs per block is not by itself enough: the blocks must also be freely combinable. The theorem strengthens the uniform baseline $3^{m/6}$ obtained from the participant-chain entropy bound, although it need not improve every degree-sensitive estimate. The proof rests on an incidence bound for labeled hypergraph matchings (Lemma 7.4); participant chains and monotone partner movement provide its application to stable marriage. The equality instances have $m = 2n$, so sharpness in m leaves the extremal rate in n open.

The participant bound also controls completions after some edges are prescribed. Let $\eta : A \rightarrow W$, for $A \subseteq M$, be a prescription extended by at least one stable matching, and put $\mathcal{S}_\eta = \{\mu \in \text{Stab}(I) : \mu|_A = \eta\}$.

Corollary 1.3 (Fixed-edge completions). *If $r = |A| < n$, then $|\mathcal{S}_\eta| < 3.3248^{n-r}$. If $r = n$, a compatible prescription has exactly one completion.*

Indeed, delete the fixed couples and restrict the preference lists. Restriction injects the original completions into stable matchings of the remaining complete instance. Deletion may introduce additional stable matchings, but this only enlarges the family used for the upper bound. The geometric-window construction also applies directly inside the conditional family, giving the following bound before the global improvement.

Theorem 1.4 (One-sided geometric bound). *For every compatible prescription $\eta : A \rightarrow W$,*

$$|\mathcal{S}_\eta| \leq C_*^{n-|A|}, \quad C_* = \exp\left(\sum_{k \geq 1} \frac{2 \log k}{(k+1)(k+2)}\right) < 3.331974.$$

The hereditary form records that the same local rate survives after conditioning on prescribed matching edges. There is also an algorithmic consequence of the improved participant count. In Popular Roommates, on a single set of participants, a matching is sought that does not lose a pairwise election against another matching, where each participant votes according to its assigned partner and treats being unmatched as worst. Kavitha’s algorithm [Kav25] uses stable-matching enumeration and has running-time base $c + 2\sqrt{c}$ when c bounds the stable-marriage count. Our rate gives $O^*(6.9717^N)$ on N participants, with polynomial factors suppressed. The counting theorem thus improves the running-time rate within an existing enumeration framework (Section 8).

1.2 A one-sided encoding

The local reduction turns a conditional choice set, governed by the full preference instance, into a one-dimensional random window on a single man’s stable-partner list. The essential structural point is that revealed edges enclose *all* feasible completions, while only the men’s coordinates incur entropy charges. Here a conditional support is simply the set of partners still possible after part of the matching has been exposed. Sample a uniform stable matching, assign the men independent uniform reveal priorities, and expose partners in increasing order. For a target u , let X_u count the partners feasible after the revealed prefix. We construct an interval of width W_u containing this support and a coupling such that, conditional on the target priority $T_u = x$,

$$X_u \leq W_u \leq L_x + R_x - 1, \quad L_x, R_x \stackrel{\text{iid}}{\sim} \text{Geom}(x).$$

Here $\text{Geom}(x)$ is the number of independent trials until the first success, each trial succeeding with probability x . The two waiting times have a law depending only on x , not on the preference instance. Their independence is local to one target; no independence between targets is needed. Thus the probabilistic envelope is uniform across instances, even though the conditional supports need not be intervals and can depend on constraints throughout the instance.

For a uniformly sampled stable matching μ , the male partner vector determines the matching and has entropy $\log |\text{Stab}(I)|$: entropy measures the average information needed to specify the random matching, using natural logarithms throughout. The chain rule expresses this entropy as a sum of conditional partner entropies along the reveal order, each bounded by the logarithm of its support size. The difficulty is to infer a useful support constraint from a revealed coordinate. If a previously revealed man is matched to a woman w on the target’s stable-partner list, her name alone does not say whether the remaining partners lie above or below w . Palmer and Pálvölgyi encode rotation ideals by the last included rotation on each of the $2n$ participant chains and use intersections with earlier chains to restrict later coordinates [PP21]. We instead encode the matching by its n male partners. A revealed coordinate then fixes an entire matching edge, and the woman’s preference between her fixed partner and the target supplies the missing direction.

To see this, recall the join property of the stable-matching lattice: giving each man the better of his partners in two stable matchings again gives a stable matching, in which every woman receives the worse of her two partners. Together with stability, this implies that the endpoints of a stable pair absent from a matching have opposite preferences between that pair and their assigned partners. Now let (u, w) be a stable pair, and suppose the revealed edge fixes w to $p \neq u$. Her comparison of

p with u is fixed throughout the conditional family, so the partner of u lies on the same side of w in every completion. Thus a revealed edge induces a fiber-wide oriented cut, while only the man's coordinate incurs an entropy charge. Validity throughout the fiber is what makes this a bound on the conditional support, rather than just a statement about the sampled matching. Independent reveal priorities then give the geometric law.

For a sampled matching μ , mark the positions on u 's stable-partner list whose owners have been revealed. A marker above $\mu(u)$ forces every feasible partner below it, and a marker below $\mu(u)$ forces every feasible partner above it. The nearest markers bracket the whole support; a missing marker is replaced by a sentinel one position beyond the list. For example, if $\mu(u) = w_4$ on a seven-partner list and the owners of w_2, w_6 are revealed, every completion assigns u one of w_3, w_4, w_5 . Other constraints may exclude some of these positions, but no feasible partner can lie outside them. We call this bracket the *revealed-pair interval*. Its width W_u bounds the support size X_u (Section 3); the possible difference between the two will matter in the global refinement. Prescribed edges provide the same cuts as edges exposed during the reveal.

Conditional on $T_u = x$, each other position on the list is marked with probability x , independently: distinct women have distinct owners in μ . Scan outward from $\mu(u)$ to the first marker on either side, extending each finite list by fresh independent Bernoulli(x) bits. Extension cannot move a first marker toward the center. The infinite waiting times are independent $L, R \sim \text{Geom}(x)$, and their window width $K_x = L + R - 1$ dominates W_u pathwise. The subtraction avoids counting the current position twice. Convolution gives $\mathbb{P}[K_x = k] = kx^2(1-x)^{k-1}$, so averaging over the target priority yields

$$\int_0^1 \mathbb{E}[\log K_x] dx = \sum_{k \geq 1} \frac{2 \log k}{(k+1)(k+2)} = \log C_*.$$

Each male coordinate therefore costs at most $\log C_*$ nats on average, proving the unconditioned C_*^n bound. For a fiber $\mathcal{S}_\eta$, reveal only the men outside A . Positions owned by men in A are deterministic markers; independently thinning each to a Bernoulli(x) marker enlarges the interval and makes the full field iid. The same rate applies to all $n - |A|$ remaining coordinates, proving Theorem 1.4. The interval law and this series already appear in the tangled-grid analysis of Palmer and Pálvölgyi [PP21, §3.4], where the rate is summed over $2n$ chain coordinates. Their final 3.55^n bound uses a further rotation-specific refinement of that analysis. Our cut lemma makes the original geometric envelope available directly on n male coordinates, yielding C_*^n .

The reduction separates a model-specific structural lemma from an instance-independent probability calculation. To reuse the argument, one needs coordinates determining the object, ordered choice lists, and revealed-coordinate cuts valid throughout each conditional family. Distinct list positions must also have distinct independently revealed owners, so that the two finite scans admit the geometric coupling. Once these conditions hold, the entropy calculation no longer uses preferences or rotations. Within stable marriage, the hereditary bound under prescribed edges is an immediate use of this separation. In a different model, the work is to establish the structural conditions, not to recompute the window distribution.

1.3 Recovering a global counting loss

The geometric window is the local relaxation; the second stage measures its unavoidable global cost. In enclosing a support by an interval and then extending the list, we forget which positions are feasible and where the real endpoints lie. To improve the resulting rate, we return to the instance and show that the local bounds cannot all be nearly tight. This does not require a better bound for every coordinate: some coordinates may have negligible loss, provided the total expected loss is

linear. Immediately before revealing u , let P_u be its conditional partner distribution. The support size X_u , interval width W_u , and infinite window K_u give three successive relaxations:

$$H(P_u) \leq \log X_u \leq \log W_u \leq \log K_u. \quad (2)$$

The three losses measure nonuniformity inside the support, infeasible partners inside the interval, and the artificial tail beyond a real list endpoint. Their sum is $\log K_u - H(P_u)$. Since $\mathbb{E} \log K_u = \log C_*$, the entropy chain gives

$$n \log C_* - \log |\text{Stab}(I)| = \mathbb{E} \sum_{u \in M} (\log K_u - H(P_u)). \quad (3)$$

Every summand is nonnegative. Our improvement recovers a linear loss from the last two terms alone. These two terms have concrete local witnesses. Call a target *shallow* if its sampled partner is at most one position from an endpoint of its stable-partner list. A short pattern of markers then makes the infinite window strictly wider than the genuine interval. Away from the endpoints, a rotation constraint can instead force the support to be a singleton while the interval still contains two positions; targets with our certificate for this event are called *extra*. In either case, averaging a bounded number of priority comparisons and dummy bits gives at least $\log 2/12$ expected slack (Lemma 6.4). The remaining task is global: prove that a typical stable matching has linearly many shallow or extra targets.

We do this by deleting exchanges from a rotation boundary. Rotations are cyclic exchanges between consecutive stable partners, with some exchanges required to precede others. Their ideals encode stable matchings: an ideal contains every predecessor of each of its elements. We use the dual orientation, in which adding a rotation improves the partners of its participating men. Rotations involving any fixed man form a chain, and his partner never returns to an earlier value. These participant constraints limit how much information can be lost when rotations are deleted from an ideal.

Let $k(D)$ count the shallow or extra targets in the matching encoded by an ideal D , and let $L(D)$ be its maximal binary rotations. A target outside those two classes has a last included rotation that is binary and maximal in D . Distinct members of $L(D)$ cannot share a man, since such rotations would be comparable. They therefore cover at least $n - k(D)$ men, two per rotation. Every subset $Q \subseteq L(D)$ can be deleted, giving a smaller ideal $J = D \setminus Q$. Thus a matching with few local witnesses has many deletion choices. To turn this into a counting loss, we need an upper bound on how many pairs (D, Q) can produce the same J .

For a fixed resulting ideal J , draw a two-colored graph on the men. Its in-edges are binary rotations maximal in J , and its out-edges are binary rotations minimal outside J . Each man lies on at most one edge of each color. Parallel in/out edges would swap the same two partners twice, contradicting monotone partner movement. Hence the graph consists of paths and even cycles of length at least four. The original boundary $L(D)$ is a matching in this graph: the out-edges identify Q , and then $D = J \cup Q$. Once its uncovered vertices are specified, every path has at most one compatible perfect matching and each intact cycle at most two.

An ideal with $k(D) \leq k$ offers at least $2^{(n-k)/2}$ deletion choices. Once its at most k uncovered vertices are specified, there are at most $2^{n/4}$ originals leading to the same J . A deletion choice uses two men, whereas a binary reconstruction ambiguity requires at least four. For sufficiently small k/n , the cost of specifying the uncovered vertices does not erase this exponential difference. Double-counting the pairs (D, Q) now shows that ideals with few chargeable targets form an exponentially small fraction of all ideals. Thus $\mathbb{E}_D k(D) \geq \alpha n$ for an absolute $\alpha > 0$ and all sufficiently large n .

The local payment (Lemma 6.4) then gives

$$\log |\text{Stab}(I)| \leq n \log C_* - \frac{\log 2}{12} \mathbb{E}_D k(D) \leq n \left(\log C_* - \frac{\alpha \log 2}{12} \right).$$

Section 6 makes the tail and α explicit, giving a base below 3.3248; replication extends the result to every size. The gain comes from combining two kinds of information. Priority events give a fixed expected loss at each witness, and deletion and reconstruction show that witnesses are abundant. The argument only averages over participants, so it requires no independence between their losses.

1.4 A sharp count in terms of stable pairs

The second proof asks how efficiently a fixed supply of stable pairs can support independent choices. It does not use the geometric bound or its global refinement. Counting all perfect matchings in its stable-pair graph gives an upper bound, but forgets both preferences and rotation precedence. Eppstein's graph-counting bound gives $6^{m/9}$, with equality for disjoint copies of $K_{3,3}$ when all perfect matchings are counted [Epp21, Lemma 6.2 of the full version]. As he observes, those perfect matchings cannot all be stable under one preference profile. Thus the graph extremizer does not settle the stable-matching extremal problem. A participant-chain application of Shearer's inequality gives the stronger baseline $3^{m/6}$, derived in Proposition 7.1. To obtain the sharp exponent, we retain not just the individual participant chains but the sets of participants shared by each rotation.

To see the baseline, let d_u be the number of stable partners of man u . The rotations involving u form a chain of length $d_u - 1$. Choose a uniformly random antichain of the rotation poset. Its projection onto this chain is either empty or contains one of those rotations, so it has at most d_u possible values. Each rotation belongs to at least two male chains. The entropy covering inequality therefore gives $|\text{Stab}(I)| \leq \prod_{u \in M} \sqrt{d_u}$. Since $d \leq 3^{d/3}$ for every positive integer d , and $\sum_u d_u = m$, the uniform rate $3^{m/6}$ follows. Keeping the degree profile can give a stronger bound on individual instances; our theorem optimizes the base of a uniform bound in m .

The additional information we retain is the labeled family of participant sets of rotations. Form a hypergraph F on the men with one labeled hyperedge for each rotation. An antichain gives a matching in F , because rotations sharing a participant are comparable. Shrinking each hyperedge to an arbitrary pair preserves every matching and produces a loopless multigraph G . With Z counting all matchings, including the empty one, the two reductions give

$$|\text{Stab}(I)| \leq Z(F) \leq Z(G).$$

Disjoint rotations can still be ordered by other precedence constraints, so the first reduction can enlarge the family. Parallel edges must keep their labels: different rotations are different choices even when their selected pairs coincide. A naive bound $Z(G) \leq 2^{(v+2e)/4}$, with v vertices and e labeled edges, fails for a component consisting of two parallel edges. That component has three matchings, whereas the proposed bound is $2^{3/2} < 3$. The labels are therefore essential to the counting problem. The correct auxiliary bound is

$$Z(G) \leq 2^{(v+2e)/4} \Gamma^{q(G)}, \quad \Gamma = \frac{3}{2\sqrt{2}},$$

where $q(G)$ counts isolated double-edge components. Its proof is an induction using the unmatched/matched recurrence at a minimum-degree vertex. Deleting vertices can create exceptional components, so the induction tracks their number in every residual graph. Equality occurs only for components on two vertices with one, two, or three parallel edges.

The remaining task is to pay for these exceptional components using the incidences lost in shrinking. Write $r = |E(F)|$, and let $\text{ex}(F) = \sum_{a \in E(F)} (|a| - 2)$ count the omitted incidences. Each touches at most two components of G : the component of the selected pair and the component of the omitted vertex. Every double-edge component of G must be touched unless it was already an isolated double-edge component of F . Hence $q(G) \leq 2\text{ex}(F)$ whenever the latter components are absent. Substitution in the graph bound yields

$$Z(F) \leq 2^{(n+2r+\text{ex}(F))/4} \beta^{\text{ex}(F)}.$$

The factor $\beta^{\text{ex}(F)}$ has a simple interpretation: the incidence budget lost in shrinking pays for all double-edge exceptions, with a strict surplus whenever a hyperedge has size greater than two. This argument does not use preferences or rotation precedence. The resulting bound (Lemma 7.4) applies to any labeled multihypergraph with edges of size at least two and no isolated component consisting of two identical size-two edges. Its budget is $B(F) = |V(F)| + \sum_{a \in E(F)} |a|$. It therefore bounds any solution family that injects into such hypergraph matchings, once this incidence budget is controlled. Both the labels and the excluded-component condition matter: labels preserve the distinct choices, and an isolated double edge violates the bound without its correction.

Rotation theory supplies exactly the missing hypothesis. An isolated component consisting of two men and two binary rotations would first swap their partners and then swap them back. No other rotation can move these men between the two swaps. This contradicts the strictly monotone progression through stable partners. The identity $m = n + \sum_{\rho} \ell(\rho) = n + 2r + \text{ex}(F)$ then gives the stable-pair bound. The induction and the application of rotation monotonicity are proved in Section 7.

We can also follow equality through the two reductions. Because $\beta < 1$, equality forces all rotations to be binary. The graph bound and the no-return property then force disjoint single-edge components covering every man. Finally every subset of rotations must be an antichain. This last condition is essential: disjoint participant sets alone do not rule out precedence. Independent copies of the two-by-two example at the start of the Introduction attain equality. By contrast, a stable-pair graph made of disjoint $K_{2,2}$'s need not attain equality; we give an explicit four-by-four example with three, rather than four, stable matchings. The theorem identifies the exact extremizers for the uniform stable-pair base. They are sparse instances, with $m = 2n$; maximizing at a fixed participant count can require much denser stable-pair graphs. Thus the two extremal questions remain distinct. Optimizing the excess penalty and describing near equality are further structural questions.

The two proofs share a specific structural restriction: a man's rotations form a chain along which his partner never returns to an earlier value. This rules out parallel in/out edges in the deletion proof and isolated double-edge components in the hypergraph application. In the first proof it limits reconstruction ambiguity; in the second it controls the cost of shrinking hyperedges. These are two counting uses of the same constraint on participant movement.

1.5 Relation to prior work

Karlin, Oveis Gharan, and Weber established the first simply exponential upper bound through mixing posets [KGW18]. Their mixing condition requires each set of rotations to meet enough participant chains. It guarantees an element with many predecessors and successors, allowing the ideal count to be bounded recursively according to whether that element is included. Our participant-count proof instead charges information to matching coordinates and controls the loss of a local relaxation. The classical lattice and rotation representation [GI89] remain common structural inputs. Every finite distributive lattice has a stable-marriage realization [GILS87]. A bound in

n must therefore use the cost and structure of its realization on $n + n$ participants, beyond the abstract ideal representation.

Palmer and Pálvölgyi’s random-reveal framework is the closest predecessor [PP21]; entropy and random ordering are shared tools [Sha48, Rad97]. Their final analysis uses both paired opposite-side chains and nearby same-side chains to improve the conditional interval. In §5 they explicitly discuss reducing $2n$ coordinates to n , while noting an obstruction for general tangled grids. Our one-sided representation uses the stable-edge comparisons available in actual matching fibers, rather than asserting such a reduction for every tangled grid. The local contribution is the fiber-wide cut and its finite coupling; the global contribution is to bound, by deletion and reconstruction, how often this envelope can be nearly tight. The classical join property supplies the stable-edge comparison, and their interval law supplies the subsequent geometric calculation.

Counting the stable matchings of a given instance raises a different complexity question. Exact counting is $\#P$ -complete even with strict complete preferences [IL86]. Chebolu, Goldberg, and Martin showed that approximate counting is equivalent, under approximation-preserving reductions, to counting independent sets in bipartite graphs ($\#BIS$), including in several low-dimensional attribute models [CGM12]. Our extremal bounds control the size of the family that these counting algorithms would have to measure. Random instances behave quite differently: when every preference list is an independent uniform permutation, the expected number of stable matchings is asymptotic to $e^{-1}n \log n$ [Pit89]. The exponential behavior of $SM(n)$ is a worst-case phenomenon.

Other work obtains exact counting or uniform sampling when the preference profile has additional structure. Bhatnagar, Greenberg, and Randall showed that the natural local walk on the stable-marriage lattice can mix exponentially slowly even in restricted preference models, so the lattice representation alone does not yield an efficient sampler [BGR08]. Cheng and Rosenbaum subsequently characterized the rotation posets arising from several parameterized preference restrictions [CR22]. In the k -range model, the Hasse diagram has pathwidth bounded as a function of k , which gives fixed-parameter algorithms for exact counting and exactly uniform sampling. For several other constant-complexity preference models, arbitrary rotation posets remain realizable and counting remains $\#BIS$ -complete. These algorithms exploit restrictions on the input to compute $|\text{Stab}(I)|$ or sample from the family. Our bounds apply to unrestricted inputs and to their conditional families, giving worst-case cardinality estimates rather than polynomial-time counting or sampling algorithms.

Prescribed edges and changes to preferences lead to a different family of algorithmic questions. Cseh and Manlove study forced and forbidden pairs, including optimization when stability and the edge restrictions cannot both be met [CM16]. Gangam, Mai, Raju, and Vazirani describe the lattice formed by stable matchings common to two nearby instances after one participant’s list changes [GMRV22]. The fixed-edge corollary here bounds the size of the exact fiber agreeing with a compatible prescription in one fixed instance.

For the stable-pair parameter, Eppstein studies which bipartite graphs can be the union of the stable matchings of one preference instance, including recognition and width-parameterized algorithms [Epp21]. His $6^{m/9}$ edge-budget bound counts all perfect matchings of such a graph [Epp21, Lemma 6.2 of the full version]. The stronger baseline $\prod_u \sqrt{d_u} \leq 3^{m/6}$ is the chain-cover consequence of Shearer’s inequality [CGFS86] proved here in Proposition 7.1. Our $2^{m/4}$ theorem uses the labeled participant sets of rotations to improve the uniform rate, and follows equality through the reduction back to the preference instance. The distinction is substantive even for disjoint $K_{2,2}$ stable-pair graphs: additional precedence can remove combinations without removing any stable pair. The equality characterization captures precisely when every combination survives. Degree-sensitive estimates retain additional information and can be stronger on individual instances.

Related matching-count bounds concern regular graphs and multigraphs [FKM08] and Hosoya

indices of simple graphs [HSX18]. We require a bound for arbitrary loopless multigraphs, retaining labeled parallel edges and the exceptional components created by deletion. The subsequent hypergraph bound separates this counting step from preferences: omitted incidences pay for the exceptions created by shrinking. Applying the tool requires its explicit excluded-component hypothesis, which rotation monotonicity verifies here.

Our algorithmic consequence combines the new count with the enumeration framework of Gusfield [Gus87] and the Popular Roommates decomposition of Kavitha [Kav25]. The improved bound on the number of stable matchings reduces the running-time base of that algorithm. A broader account of structural, counting, and algorithmic questions in matching under preferences appears in [GI89, CCM19]. Sections 2 to 5 establish the conditional geometric bound; Section 6 proves the global improvement. Sections 7 and 8 give the stable-pair bound and algorithmic consequence. The appendices contain coupling details and exact certificates.

2 Stable matchings, fibers, and random reveals

Fix a strict, complete stable-marriage instance $I = (M, W, \succ)$ with n men and n women. We write $w \succ_u w'$ when man u prefers w to w' , and $u \succ_w u'$ for the analogous comparison by a woman. A perfect matching is a bijection $\mu : M \rightarrow W$; the partner of w is $\mu^{-1}(w)$. A pair (u, w) blocks μ if $w \succ_u \mu(u)$ and $u \succ_w \mu^{-1}(w)$. The matching is *stable* if it has no blocking pair, and $\text{Stab}(I)$ denotes the set of stable matchings of I . A pair (u, w) is a *stable pair* if it belongs to some stable matching. The *stable-partner list* of u contains all such women, ordered from best to worst by u . We use the lattice structure of stable matchings in the interval argument only through the join property stated in Proposition 3.1.

Revealing men's partners one at a time leaves a family of stable matchings consistent with the exposed edges.

Definition 2.1 (Stable fiber and conditional partner support). For $\mu \in \text{Stab}(I)$, a revealed set $A \subseteq M$, and a target $u \notin A$, the *stable fiber* and *conditional partner support* are

$$\begin{aligned}\mathcal{F}(\mu, A) &= \{\nu \in \text{Stab}(I) : \nu|_A = \mu|_A\}, \\ \mathcal{X}_u(\mu, A) &= \{\nu(u) : \nu \in \mathcal{F}(\mu, A)\}.\end{aligned}$$

Write $X_u(\mu, A) = |\mathcal{X}_u(\mu, A)|$ for the support size.

The sampled matching μ belongs to its fiber, so these sets are nonempty. The projection $\mathcal{X}_u(\mu, A)$ lists the possible partners of the next man; its size will bound his conditional entropy. We use the entropy chain rule and the support bound $H(Y) \leq \log |\text{supp}(Y)|$. For stable matchings sampled uniformly, the male partner vector determines the matching, and its conditional supports are exactly the projections $\mathcal{X}_u(\mu, A)$; see Lemma 4.1. All logarithms are natural.

The support bound depends on the reveal order. We average over this order by assigning independent priorities $T_p \sim \text{Unif}[0, 1]$ to the men and sorting them increasingly. A fixed ordering of the names breaks ties. For a target u , the earlier men are $A_u(T) = \{p \in M : (T_p, p) < (T_u, u)\}$. When analyzing one target, we fix $T_u = x$, average over the other priorities, and then integrate over $x \in [0, 1]$. At fixed x , the searches for earlier men will lead to geometric waiting times. Our convention is that, for $0 < x \leq 1$, $G \sim \text{Geom}(x)$ means $\mathbb{P}[G = r] = x(1 - x)^{r-1}$ for $r = 1, 2, \dots$

3 The revealed-pair interval

A revealed edge constrains every stable completion of a prefix. If its female endpoint is a stable partner of another man, it fixes which side of that endpoint his partner must lie on. The nearest constraints on the two sides enclose his conditional support. We derive this from the classical men-better join.

Proposition 3.1 (Men’s join). *Let μ and σ be stable matchings. Give each man the partner he prefers between his partners in μ and σ . These choices form a stable perfect matching λ . Every woman receives the partner she likes less between her partners in μ and σ .*

This is the standard join operation in the stable-matching lattice. We use only its participantwise formulation above; see [Knu76, GI89]. To extract a constraint from a revealed edge, we first compare the two endpoints of a stable pair that is absent from the current matching. Stability rules out mutual improvement; the join rules out mutual worsening.

Lemma 3.2 (Stable-pair sidedness). *Suppose (u, w) is a stable pair and μ is a stable matching not containing it. Exactly one of the following alternatives holds:*

$$\begin{aligned} \mu(u) \succ_u w \quad \text{and} \quad u \succ_w \mu^{-1}(w), \\ w \succ_u \mu(u) \quad \text{and} \quad \mu^{-1}(w) \succ_w u. \end{aligned}$$

Proof. Choose a stable matching σ containing (u, w) . It is impossible that u prefers $\mu(u)$ to w while w prefers $\mu^{-1}(w)$ to u : the men’s join of μ and σ would assign $\mu(u)$ to u , whereas the women-worse rule would assign u to w , contradicting Proposition 3.1. It is also impossible that both u and w prefer one another to their partners in μ , because then (u, w) blocks μ . Strict preferences exclude ties, so only the two opposite-sided alternatives remain. $\square$

Thus, once a woman’s partner is fixed, her comparison with a target man cannot change across completions. The opposite comparison on his list cannot change either.

Corollary 3.3 (A fixed endpoint partner fixes the side). *Let (u, w) be a stable pair. If stable matchings μ, ν give w the same partner $p \neq u$, then $\mu(u)$ and $\nu(u)$ lie on the same side of w in u ’s preference order.*

Proof. The comparison of p and u in w ’s strict order is the same in the two matchings. By Lemma 3.2, the comparison on u ’s side must be the opposite one in both matchings. $\square$

Apply this comparison inside the stable fiber of Definition 2.1. Each revealed edge gives a one-sided constraint, and the nearest constraints on the two sides contain all feasible partners in a single interval. The following statement also covers missing constraints: the indices 0 and $q + 1$ are sentinels, not women.

Lemma 3.4 (Revealed-pair interval). *Fix a stable matching μ , a target man u , and a revealed set $A \subseteq M \setminus \{u\}$. List the stable partners of u from best to worst as*

$$w_1 \succ_u w_2 \succ_u \cdots \succ_u w_q, \quad \mu(u) = w_j.$$

Let a be the largest $h < j$ such that $w_h = \mu(p)$ for some $p \in A$, or $a = 0$ if no such index exists. Let b be the smallest $h > j$ with this property, or $b = q + 1$ if none exists. Then every $\nu \in \mathcal{F}(\mu, A)$ satisfies

$$a < \text{rank}_u(\nu(u)) < b.$$

Consequently, $|\mathcal{X}_u(\mu, A)| \leq b - a - 1$.

Proof. Consider a revealed index $h \neq j$, with $w_h = \mu(p)$ for $p \in A$. Since $\nu \in \mathcal{F}(\mu, A)$, both μ and ν give w_h the same partner p . Moreover $p \neq u$, because $h \neq j$ and μ is a bijection. If $h < j$, then u prefers w_h to $\mu(u)$. Corollary 3.3 forces $\nu(u)$ to lie on the same, worse side of w_h , so $\text{rank}_u(\nu(u)) > h$. If $h > j$, the symmetric argument gives $\text{rank}_u(\nu(u)) < h$. Taking the nearest revealed index on each side gives the strict interval. It contains exactly $b - a - 1$ integer indices, and the partner list has no duplicates. $\square$

The interval contains the partners in every completion of the revealed edges. We next apply this bound along a random reveal order.

4 The one-sided entropy encoding

Apply the interval bound along a random reveal order. Fix an instance I , write $\mathcal{S} = \text{Stab}(I)$, and choose μ uniformly from $\mathcal{S}$. Independently sample the priority vector T from Section 2. Immediately before revealing u , the support size is $X_u(\mu, T) = |\mathcal{X}_u(\mu, A_u(T))|$.

Lemma 4.1 (Entropy charged to men). *For every fixed reveal order, the entropy chain is bounded by the sum of the logarithms of the corresponding prefix-fiber projections. Consequently,*

$$\log |\mathcal{S}| \leq \mathbb{E}_{\mu, T} \sum_{u \in M} \log X_u. \quad (4)$$

Proof. The vector $(\mu(u))_{u \in M}$ determines the matching and is uniform on $\mathcal{S}$. The entropy chain rule therefore writes $\log |\mathcal{S}|$ as the sum of the conditional entropies of the male partner coordinates. At a realized prefix, the next coordinate is supported on exactly the partners attained by stable matchings agreeing with that prefix, namely the projection $\mathcal{X}_u$ for the man being revealed. Bounding conditional entropy by the logarithm of support size proves the fixed-order inequality. Averaging over the independent priority order proves (4). $\square$

It remains to bound one summand in (4). Fix (μ, T) and a target u , and write its stable-partner list as $w_1 \succ_u \cdots \succ_u w_q$, with $\mu(u) = w_j$. For $h \neq j$, let $p_h = \mu^{-1}(w_h)$. We mark position h when $p_h \in A_u(T)$. Let $a < j < b$ be the nearest marked positions, with sentinels $a = 0$ and $b = q + 1$ when the corresponding side has no marker. The revealed-pair interval gives

$$X_u(\mu, T) \leq b - a - 1. \quad (5)$$

For the averaging step, it is more convenient to measure from the current partner than from the list endpoints. Let $\Delta_u^L = j - a$ and $\Delta_u^R = b - j$ be the one-based distances from the current partner to the first marker on the two sides, with the sentinels counted as one position beyond the list. The window has width $W_u = \Delta_u^L + \Delta_u^R - 1$, and (5) states $X_u \leq W_u$. These distances depend only on which owners precede the target. Their distribution under random priorities gives the bound on $\mathbb{E} \log X_u$ in the next section; Section B records the sentinel cases.

5 The geometric rate and the hereditary theorem

Conditional on the target's priority, the marker bits are independent: distinct positions have distinct owners in the sampled matching. Extending the two finite marker lists by independent tails bounds their cut distances by geometric waiting times.

Lemma 5.1 (Distinct marker sources). *For fixed μ, u , the men $p_h = \mu^{-1}(w_h)$, $h \neq j$, are pairwise distinct and none is u .*

Proof. The women w_h are distinct and μ^{-1} is a bijection, so their matched men are distinct. Moreover, $p_h = u$ would imply $w_h = \mu(u) = w_j$, contrary to $h \neq j$. $\square$

Distinct owners let us analyze the two sides together. Fix the target priority at $T_u = x \in (0, 1]$. Every position $h \neq j$ is marked precisely when $T_{p_h} < x$. By Lemma 5.1, the complete left-right marker vector is iid Bernoulli(x): its entries are threshold events for distinct independent priorities, and ties occur with probability zero. Couple the two finite marker lists to prefixes of independent infinite Bernoulli(x) streams, and let L, R be their first-success locations. Extending a list cannot move its first success toward the center, so the finite cut distances are at most L, R .

Lemma 5.2 (Geometric-window domination). *For every fixed matching μ , target u , and priority $T_u = x \in (0, 1]$, there is a coupling under which*

$$X_u \leq W_u \leq L + R - 1, \quad L, R \stackrel{\text{ind}}{\sim} \text{Geom}(x).$$

Proof. The common-prefix coupling just described gives $\Delta_u^L \leq L$ and $\Delta_u^R \leq R$. Combine these inequalities with the two-sided window bound at the end of Section 4. A pathwise construction covering the sentinel cases appears in Section B. $\square$

It remains to average the logarithm of this window over the target priority. Put $K_x = L + R - 1$. Convolution gives

$$\mathbb{P}[K_x = k] = kx^2(1-x)^{k-1}, \quad k \geq 1. \quad (6)$$

Lemma 5.3 (One-coordinate entropy rate). *For every stable-marriage instance and every man u ,*

$$\mathbb{E} \log X_u \leq S_*, \quad S_* = \sum_{k \geq 1} \frac{2 \log k}{(k+1)(k+2)}.$$

Proof. Condition on $T_u = x$, apply Lemma 5.2, and then integrate over x . Tonelli's theorem and the beta integral give

$$\begin{aligned} \mathbb{E} \log X_u &\leq \int_0^1 \sum_{k \geq 1} (\log k) k x^2 (1-x)^{k-1} dx \\ &= \sum_{k \geq 1} \frac{2 \log k}{(k+1)(k+2)} = S_*. \end{aligned}$$

The terms are nonnegative, so Tonelli applies; the resulting series is finite because its summand is $O((\log k)/k^2)$. $\square$

Summing over the men gives the analytic bound. For a compatible edge prescription, the additional markers are deterministic. Thinning them to fresh Bernoulli bits enlarges the interval, so the same rate applies to each remaining coordinate.

Proof of Theorem 1.4. Put $U = M \setminus A$, choose μ uniformly from $\mathcal{S}_\eta$, and reveal only the coordinates in U . The entropy chain gives $\log |\mathcal{S}_\eta| \leq \sum_{u \in U} \mathbb{E} \log X_u^\eta$, where X_u^η is the current partner support inside the conditional family. Fix a target priority x . Matched men in U again supply independent Bernoulli(x) markers, whereas each matched man in A supplies a deterministic marker. Replace every deterministic marker by a fresh independent Bernoulli(x) bit and keep the threshold bit at

every other position. Bijectivity makes the resulting full field iid Bernoulli(x). Since this thinning only removes cuts, its interval contains the genuine revealed-pair interval. The same geometric domination therefore gives $\mathbb{E} \log X_u^\eta \leq S_*$ for each $u \in U$. Consequently,

$$\log |\mathcal{S}_\eta| \leq (n - |A|)S_*,$$

and exponentiation proves the theorem. The coordinatewise thinning construction and the case $U = \emptyset$ are detailed in Section A. $\square$

Corollary 5.4 (Analytic bound). *Every n -by- n instance I satisfies*

$$|\text{Stab}(I)| \leq \exp(nS_*) = C_*^n.$$

Consequently $\text{SM}(n) \leq C_*^n$.

Proof. Take $A = \emptyset$ in Theorem 1.4 and then maximize over instances. For $n = 0$, both sides are one. $\square$

6 Recovering slack at rotation boundaries

The geometric window gives the same upper bound for every coordinate, regardless of the length of its partner list or the precedence constraints among rotations. We now use these two sources of information to improve the bound by a fixed amount per participant.

Theorem 6.1 (Linear slack in the geometric domination). *For every stable-marriage instance with $n \geq 850001$,*

$$\log |\text{Stab}(I)| \leq n(S_* - \delta_{\text{ch}}), \quad \delta_{\text{ch}} = \frac{616029 \log 2}{196608000}.$$

The proof has a local and a global part. Locally, a target close to a list endpoint, or with an additional rotation constraint, contributes at least $\log 2/12$ expected slack. Globally, an ideal with few such targets must have many disjoint binary rotations on its boundary. Deleting subsets of this boundary creates more choices than can be lost when reconstructing the ideal: a deletion choice uses two men, whereas a binary reconstruction ambiguity requires an alternating cycle on at least four. This yields an exponential tail bound for the number of chargeable targets and hence the required average payment.

We prove the local witnesses and the deletion bound below. The dual rotation convention is recorded in Section D.1; only the elementary tail estimates are deferred to Section D.2. The final conversion from $S_* - \delta_{\text{ch}}$ to $\log 3.3248$, including removal of the size threshold, is given in Section 6.3.

6.1 Local witnesses and their payment

Fix an instance with stable set $\mathcal{S}$. At a reveal node for a target u , let P_u be the conditional partner distribution, X_u its support size, W_u the width of the revealed-pair interval, and K_u the coupled infinite geometric window. The inequalities $H(P_u) \leq \log X_u \leq \log W_u \leq \log K_u$ decompose the loss in the fiber theorem into three nonnegative terms:

$$\begin{aligned} nS_* - \log |\mathcal{S}| &= \mathbb{E} \sum_{u \in M} (\log K_u - H(P_u)), \\ \log K_u - H(P_u) &= (\log X_u - H(P_u)) + \log \frac{W_u}{X_u} + \log \frac{K_u}{W_u}. \end{aligned} \tag{7}$$

The terms measure nonuniformity, support slack, and finite-window slack. All are nonnegative. The refinement uses only the last two. Indeed, the entropy chain rule gives $\log |\mathcal{S}| = \mathbb{E} \sum_u H(P_u)$, while the infinite-stream calculation in Lemma 5.3 gives $\mathbb{E} \log K_u = S_*$ for every target.

To lower-bound the last two terms, we distinguish targets near a list endpoint from those with an additional rotation constraint. Represent stable matchings by ideals D of the dual rotation poset: reverse the usual elimination order and complement the eliminated set, so adding rotations improves the men's partners. List the stable partners of a man u from best to worst as $w_0(u), \dots, w_{q_u-1}(u)$, and suppose that the matching of D assigns $w_j(u)$ to u . If C_u is the chain of rotations containing u , then $j = q_u - 1 - |D \cap C_u|$. Thus fixing a partner fixes the cut of its participant chain; Section D.1 records the full dictionary.

Definition 6.2 (Chargeable targets). With the dual orientation and indexing above, call (D, u) *shallow* when $\min\{j, q_u - 1 - j\} \leq 1$. Otherwise let $\rho(D, u)$ be the last included rotation on the u -chain. Call (D, u) *extra* when $\rho(D, u)$ either contains at least three men or is not maximal in D . Let $k(D)$ count the shallow and extra targets. By definition, only nonshallow targets are classified as extra, so these two classes are disjoint. We call a target *chargeable* if it belongs to either class.

The two kinds of targets expose the two ways in which the infinite window overcounts. A shallow target reaches the end of its finite partner list. An extra target has a second revealed edge that can force the relevant rotation cut even though the marker interval still contains two positions. The next lemma makes the latter witness explicit. All expectations in this subsection include the reveal priorities and the independent dummy Bernoulli tails used by the common-prefix coupling.

Lemma 6.3 (Witness for an extra target). *For every extra (D, u) , there is a man p , distinct from u and from the matched man of $w_{j+1}(u)$, such that fixing p 's partner to its value in the matching of D forces $\rho(D, u)$ to remain included. Consequently, every matching in that one-coordinate fiber assigns u a partner of index at most j .*

Proof. Write $\rho = \rho(D, u)$. If ρ contains at least three men, choose a participant p other than u and the matched man of $w_{j+1}(u)$. By Lemma D.1, fixing p 's partner fixes the cut of its rotation chain, so every ideal in the resulting fiber contains ρ .

Otherwise ρ has length two and is not maximal. Choose $\sigma \in D$ with $\rho < \sigma$. The rotation σ does not contain u , because ρ is the last included rotation on the u -chain. It contains at least two men, so it has a participant p different from the matched man of $w_{j+1}(u)$. By Lemma D.1, fixing p 's partner forces σ to remain included, and ideal closure then forces ρ . In either case the cut of the u -chain cannot move past ρ , which is exactly the stated bound on the partner index. $\square$

Lemma 6.4 (Local payment). *Conditional on the base ideal D , the total expected support and finite-window slack satisfies*

$$\mathbb{E} \sum_{u \in M} [(\log W_u - \log X_u) + (\log K_u - \log W_u)] \geq \frac{\log 2}{12} k(D). \quad (8)$$

Proof. Fix a target priority $T_u = x$. If (D, u) is shallow, choose a *short* side containing at most one genuine position and call the other side the *opposite* side. We use three events in the two independent infinite Bernoulli streams. In the event descriptions, bit indices start at zero; a first success at index r has geometric waiting time $r + 1$.

- (i) The first three short-side bits fail and the first opposite-side bit succeeds. Then $W_u \leq 2$, $K_u \geq 4$, so the finite-window slack is at least $\log 2$. The conditional mass is $x(1 - x)^3$.

- (ii) The first short-side success occurs at index two and the first opposite-side success at index zero. Then $W_u \leq 2$, $K_u = 3$, so the slack is at least $\log(3/2)$. The conditional mass is $x^2(1-x)^2$.
- (iii) The first three short-side bits fail and the first opposite-side success occurs at index one. Then $W_u \leq 3$, $K_u \geq 5$, so the slack is at least $\log(5/3)$. The conditional mass is $x(1-x)^4$.

The events are pairwise disjoint: the first two distinguish a short-side first-success index of at least three from exactly two, and the first and third distinguish opposite-side first-success indices zero and one. Integrating their payments over x gives

$$\frac{\log 2}{20} + \frac{\log(3/2)}{30} + \frac{\log(5/3)}{30} \geq \frac{\log 2}{12}. \quad (9)$$

Indeed, the three beta integrals are respectively $1/20, 1/30, 1/30$, and the inequality follows from $\log(3/2) + \log(5/3) = \log(5/2) \geq \log 2$.

Now suppose (D, u) is extra (and hence is not shallow). Let p be the witness from Lemma 6.3, and let p_- and p_+ be matched to the immediate left and right stable partners of u . The men u, p_-, p_+ are pairwise distinct; the witness p is distinct from u, p_+ but may equal p_- . If $p \neq p_-$, set $s = p_-$; otherwise choose any man $s \notin \{u, p, p_+\}$. A nonshallow target has at least two stable partners on each side of its current partner, so $q_u \geq 5$ and hence $n \geq 5$. The required s therefore exists. The three men p, s, p_+ are distinct and different from u . The event $T_p < x, T_s < x < T_{p_+}$ has conditional probability $x^2(1-x)$. Both p and p_- are then revealed. The edge of p forces the target index to be at most j , while the revealed edge of p_- forces it to be greater than $j-1$. Thus $X_u = 1$. The matched man p_+ is unrevealed, so the marker interval still contains $w_j(u)$ and $w_{j+1}(u)$; hence $W_u \geq 2$, and the support slack is at least $\log 2$.

All constrained priority coordinates are distinct, so integration gives an expected payment of $\log 2/12$, since $\int_0^1 x^2(1-x) dx = 1/12$. Use (9) for a shallow target; if the target is not shallow but is extra, use the support-gap event above. The two classes cover exactly the targets counted by $k(D)$ and are disjoint. Summing proves the lemma. $\square$

6.2 Counting ideals with few chargeable targets

The local bound is useful only if a typical ideal has many chargeable targets in the sense of Definition 6.2. We prove this by encoding ideals with few such targets through deletions from their rotation boundary. First, every uncharged target must lie on a maximal binary rotation. Let $L(D)$ be the length-two rotations maximal in D . Rotations that share a man form a chain, so the members of $L(D)$ are pairwise disjoint on men. For a target that is neither shallow nor extra, the last included rotation exists, has length two, and is maximal in D , hence belongs to $L(D)$. Each member of $L(D)$ covers at most its two participating men. Consequently,

$$|L(D)| \geq \frac{n - k(D)}{2}. \quad (10)$$

Each subset of this boundary can be deleted independently. To turn these many deletion choices into a bound on the number of original ideals, we must control how many choices can lead to the same image.

Definition 6.5 (Rotation boundary graph). For an ideal J of the dual rotation poset, the *boundary graph* G_J is the two-colored graph on the men whose edges are labeled by binary rotations and join their two male participants. Its in-edges are length-two rotations maximal in J , and its out-edges are length-two rotations minimal outside J .

This graph controls how many different pairs (D, Q) , with $Q \subseteq L(D)$, can produce the same deleted ideal $J = D \setminus Q$.

Lemma 6.6 (Boundary graph). *The graph G_J is simple and has maximum degree two. At every degree-two vertex its incident edges have different colors. Thus every component is a path or an even cycle of length at least four. The number of matchings in G_J leaving at most k vertices unmatched is at most*

$$2^{n/4} \sum_{i=0}^k \binom{n}{i}.$$

Proof. A participant chain contains at most one rotation maximal in J and at most one rotation minimal outside J . This proves the degree bound and the alternation of colors. Suppose an in-edge ρ and an out-edge σ used the same two men. Maximality of ρ in J and minimality of σ outside J make them consecutive on the rotation chain of each of those men. Indeed, ideal closure forces $\rho < \sigma$. A rotation strictly between them would contradict maximality of ρ if it belonged to J , and minimality of σ outside J otherwise. Thus neither man changes partners between ρ and σ . A length-two rotation exchanges the two women currently matched to its two men, so the later of ρ and σ would exchange the same two women a second time. One of the men would then return to an earlier partner, contradicting the strict monotone movement along his participant chain. Hence the graph is simple, and its cycles are even and have length at least four.

Fix the set U of unmatched vertices. Every path component of $G_J - U$ has at most one perfect matching, while an even cycle has at most two. The cycles are vertex-disjoint and use at least four vertices each, so there are at most $n/4$ of them. Summing the resulting $2^{n/4}$ bound over the possible sets U proves the displayed estimate. $\square$

Each binary rotation uses two men and offers a deletion choice, whereas a cycle causing a binary reconstruction ambiguity uses at least four. Double-counting ideals together with their deletion choices gives the tail bound below.

Lemma 6.7 (Boundary-deletion tail). *If D is a uniformly random rotation ideal, then for every integer $0 \leq k \leq n$,*

$$\mathbb{P}[k(D) \leq k] \leq 2^{-(n-k)/2+n/4} \sum_{i=0}^k \binom{n}{i}. \quad (11)$$

Proof. Let $\mathcal{B}_k = \{D : k(D) \leq k\}$, and count pairs (D, Q) with $D \in \mathcal{B}_k$ and $Q \subseteq L(D)$. By (10), there are at least $|\mathcal{B}_k| 2^{(n-k)/2}$ pairs. Map each pair to $J = D \setminus Q$. The elements of Q are pairwise incomparable maximal elements of D , so removing any subset of them preserves the ideal property. If $\rho \in L(D) \setminus Q$, no element above ρ belongs to D , and deletion cannot create one; hence ρ remains maximal in J and gives an in-edge of G_J . If $\rho \in Q$, every predecessor of ρ lies in D . No such predecessor belongs to Q , because Q consists of incomparable maximal elements. All predecessors therefore remain in J , so ρ is minimal outside J and gives an out-edge.

The union of these boundary edges is a matching because the rotations in $L(D)$ are pairwise disjoint on men; by (10), it leaves at most k men unmatched. For fixed J , its colors uniquely recover the deletion: the out-edges are exactly Q , the in-edges are exactly $L(D) \setminus Q$, and then $D = J \cup Q$. Thus Lemma 6.6 bounds the number of preimages of every J . Divide by the total number of ideals to obtain (11). $\square$

The tail bound converts the local payment into a linear average loss. We choose a rational threshold strictly inside its exponentially decaying range so that the remaining comparison can be checked exactly. Set $\alpha = 47/1250$, $k = \lfloor \alpha n \rfloor$, and $h(t) = -t \log t - (1-t) \log(1-t)$. The standard

binomial-entropy estimate and the exact comparison in Section D.2 give

$$\sum_{i=0}^k \binom{n}{i} \leq e^{nh(\alpha)}, \quad \left(\frac{1}{4} - \frac{\alpha}{2}\right) \log 2 - h(\alpha) > \frac{\log 2}{50000}.$$

Substitution in Lemma 6.7 yields

$$\mathbb{P}[k(D) \leq \lfloor \alpha n \rfloor] \leq \exp\left(-\frac{n \log 2}{50000}\right) < 2^{-16}$$

for $n \geq 850001$. On the complementary event, integrality gives $k(D) > \alpha n$. We have proved the following average bound.

Lemma 6.8 (Linear average charge). *For every $n \geq 850001$,*

$$\mathbb{E}_D k(D) > \frac{616029}{16384000} n. \quad (12)$$

Proof. The preceding tail estimate gives

$$\mathbb{E}_D k(D) > (1 - 2^{-16}) \alpha n = \frac{616029}{16384000} n.$$

□

Combining the entropy identity with the local payment proves the theorem.

Proof of Theorem 6.1. Under the rotation-ideal bijection, a uniform stable matching gives a uniform ideal D . Drop the nonuniformity term from (7), apply Lemma 6.4, and average Lemma 6.8. This gives

$$nS_* - \log |\text{Stab}(I)| > \frac{\log 2}{12} \frac{616029}{16384000} n = \delta_{\text{ch}} n,$$

which is stronger than the displayed weak inequality. □

6.3 Exact evaluation and removal of the size threshold

The structural argument gives an exponent for sufficiently large instances. Two steps turn it into the stated bound for every size: an exact numerical comparison, followed by replication. The numerical step also evaluates the unrefined geometric rate used in the hereditary theorem.

Proposition 6.9 (Evaluation of the rates).

$$C_* < 3.331974, \quad \exp(S_* - \delta_{\text{ch}}) < 3.3248. \quad (13)$$

Proof. Abel summation, a telescoping tail, and positive logarithm expansions reduce both comparisons to strict rational inequalities. The complete calculation is in Section C. □

The size threshold costs nothing in the final bound. A small instance violating it could be replicated into arbitrarily large instances with the same exponential rate, as the following standard argument shows.

Lemma 6.10 (Replication). *If $c > 1$ and $\text{SM}(N) < c^N$ for every $N \geq N_0$, then $\text{SM}(n) < c^n$ for every $n \geq 1$.*

Proof. Fix an instance I of size $n + n$, with s stable matchings. Take t disjoint copies, putting all within-copy partners above all outside partners. Every independent selection of stable matchings in the copies is stable in the enlarged complete instance: no outside pair can block, and no within-copy pair blocks its selected matching. Thus $s^t \leq \text{SM}(tn)$. Choose t with $tn \geq N_0$. Then $s^t < c^{tn}$, and taking t -th roots gives $s < c^n$. $\square$

Proof of Theorem 1.1. For $n \geq 850001$, Theorem 6.1 and the second comparison in (13) give $\text{SM}(n) < (2078/625)^n$. Apply Lemma 6.10. The first comparison also supplies the stated numerical value of the geometric bound in Theorem 1.4. $\square$

Finally, the refined bound passes to prescribed-edge completions by an injection into a smaller instance.

Proof of Corollary 1.3. Delete the r prescribed couples and restrict the strict complete preference lists to the remaining $n - r$ men and women. Every stable completion restricts to a stable matching of this smaller instance: any blocking pair there would also block the original completion. The restriction map is injective because all deleted edges are fixed. If $n - r \geq 1$, apply Theorem 1.1; if $n = r$, the prescribed edges already determine the matching. $\square$

7 Counting by stable pairs

We turn from a bound per participant to the sharp bound per stable pair in Theorem 1.2. The proof does not use the preceding entropy improvement. Instead, it retains participant conflicts between rotations, bounds the resulting hypergraph matchings, and uses monotone partner movement to exclude the exceptional components. The argument separates into a general counting theorem and a model-specific representation: the multigraph and hypergraph lemmas use only labeled edges and incidences; the application in Section 7.4 uses rotation theory to verify their hypotheses and identify the incidence budget with m . Write d_u for the number of stable partners of participant u , and m for the number of distinct stable pairs.

7.1 Participant chains and a baseline

We use standard rotation theory as reviewed in [KGW18, §4] and [Epp21, §2]. Stable matchings correspond bijectively to ideals of the rotation poset. For a participant u with d_u stable partners, the rotations involving u form a chain of length $d_u - 1$. Eliminating all rotations in a linear extension moves each man's partner strictly downward through his stable partners. A rotation has length at least two; a binary rotation swaps the current partners of its two men. Summing the chain lengths over the men gives

$$m = \sum_{u \in M} d_u = n + \sum_{\rho \in \mathcal{R}(I)} \ell(\rho). \quad (14)$$

For any finite poset, an ideal is uniquely determined by its maximal elements, and every antichain generates an ideal. Thus the number of stable matchings is also the number of antichains of $\mathcal{R}(I)$. Covering these antichains by participant chains gives a useful baseline, but not yet the sharp stable-pair exponent.

Proposition 7.1 (Participant-chain bound). *For either side $V \in \{M, W\}$,*

$$|\text{Stab}(I)| \leq \prod_{u \in V} \sqrt{d_u} \leq 3^{m/6}.$$

Proof. Choose a uniformly random antichain A of the rotation poset and regard it as a zero-one vector indexed by rotations. For each $u \in V$, let C_u be the chain of rotations involving u . Its projection has at most d_u possible values: the empty set or one of its $d_u - 1$ elements. Every rotation coordinate occurs in at least two chains. Shearer's inequality [CGFS86] therefore gives

$$\log |\text{Stab}(I)| = H(A) \leq \frac{1}{2} \sum_{u \in V} H(A|_{C_u}) \leq \frac{1}{2} \sum_{u \in V} \log d_u.$$

Here $A|_{C_u}$ is the restriction of the zero-one vector to C_u . For integers $d \geq 1$, $d \leq 3^{d/3}$: check $d = 1, 2, 3$, then use $(d+1)/d \leq 4/3 < 3^{1/3}$ for $d \geq 3$. Finally $\sum_{u \in V} d_u = m$. $\square$

7.2 A multigraph bound with exceptional components

An antichain of rotations uses disjoint sets of men and therefore gives a hypergraph matching. We bound these matchings by shrinking each hyperedge to a pair and applying a multigraph estimate. Related matching-count bounds treat regular multigraphs [FKM08] and simple graphs [HSX18]. We use the following self-contained induction for arbitrary loopless multigraphs, retaining the exceptional components created by vertex deletion, even when they are absent from the original graph. All multigraphs in this section are finite and loopless. Parallel edges are distinct labeled objects. The number $Z(G)$ counts *all* matchings, including the empty matching. Write $v = |V(G)|$, let e count edges with multiplicity, and let $q(G)$ count components consisting of exactly two vertices joined by exactly two edges. Set $\Gamma = 3/(2\sqrt{2}) > 1$; the ratio $\Gamma/\sqrt{2} = 3/4$ will control the induction estimates.

Lemma 7.2 (Multigraph matching bound). *Every such multigraph satisfies*

$$Z(G) \leq 2^{(v+2e)/4} \Gamma^{q(G)}. \quad (15)$$

Equality holds exactly when every component consists of two vertices joined by one, two, or three parallel edges. The empty graph is included.

Proof. Induct on v . Both sides multiply over components, so it suffices to handle connected graphs. One isolated vertex gives strict inequality. A two-vertex component with $k \geq 1$ edges has $Z = k + 1$. For $k = 1, 3$ the uncorrected bound is exact, and for $k = 2$ the factor Γ makes it exact. For $k \geq 4$, strictness follows from equality at $k = 3$ and $(k+2)/(k+1) \leq 5/4 < \sqrt{2}$.

Suppose now that G is connected and $v \geq 3$, so $q(G) = 0$. Choose x with minimum weighted degree d . For a neighbor y , put $D = \deg(y) \geq d$ and let $a = a_{xy}$ be the multiplicity of xy . The exact recurrence is

$$Z(G) = Z(G - x) + \sum_{y \sim x} a_{xy} Z(G - x - y). \quad (16)$$

Let $q_x = q(G - x)$ and $q_{xy} = q(G - x - y)$. Every double-edge component created by a deletion has an edge to the deleted set in G , since G was connected. Different new components use distinct crossing edges. Hence

$$q_x \leq d, \quad q_{xy} \leq d + D - 2a = |\delta(\{x, y\})|. \quad (17)$$

The number of edges *deleted* in the second graph is $d + D - a$.

Applying induction in (16) and dividing by $2^{(v+2e)/4}$, the unmatched term is at most

$$U = 2^{-1/4 - d/2} \Gamma^{q_x}, \quad (18)$$

and the contribution of neighbor y is at most

$$T_y = a 2^{-1/2 - (d+D-a)/2} \Gamma^{q_{xy}}. \quad (19)$$

It suffices to show $U + \sum_y T_y < 1$. We distinguish the possible minimum degrees.

Case $d = 1$. There is one neighbor, with $a = 1$ and $D \geq 2$. If $D = 2$, then $q_x = 0$: a new component must contain y , whose remaining degree is one. Also $q_{xy} \leq 1$, giving

$$U + T_y \leq 2^{-3/4} + \frac{3}{8} < \frac{5}{8} + \frac{3}{8} = 1.$$

If $D \geq 3$, then $q_x \leq 1$. The upper bound $2^{-1/2-D/2}\Gamma^{D-1}$ for T_y decreases with D by ratio $3/4$; its value at $D = 3$ is $9/32$. Therefore

$$U + T_y \leq 2^{-3/4}\Gamma + \frac{9}{32} < \frac{2}{3} + \frac{9}{32} < 1.$$

Case $d = 2$, with two neighbors. Both incident edges have multiplicity one. Let $t \in \{0, 1, 2\}$ be the number of neighbors of weighted degree at least three. A degree-two neighbor has degree one in $G - x$, so $q_x \leq t$. For $D = 2$, (17)–(19) give $T_y \leq \Gamma^2/4 = 9/32$. For $D \geq 3$, they give

$$T_y \leq 2^{-1/2-(D+1)/2}\Gamma^D \leq \frac{27}{128},$$

again by the ratio $3/4$. The three sums are bounded as follows:

$$\begin{aligned} t = 0 : \quad & 2^{-5/4} + 9/16 < 7/16 + 9/16 = 1, \\ t = 1 : \quad & 2^{-5/4}\Gamma + 63/128 < 1/2 + 63/128 < 1, \\ t = 2 : \quad & 2^{-5/4}\Gamma^2 + 54/128 < 1/2 + 54/128 < 1. \end{aligned}$$

Case $d = 2$, with one neighbor. Here $a = 2$, $D \geq 3$, and $q_x \leq 1$. The bound for T_y decreases with D and equals $\Gamma/2$ at $D = 3$. Thus

$$U + T_y \leq 2^{-5/4}\Gamma + \frac{\Gamma}{2} < \frac{15}{32} + \frac{17}{32} = 1.$$

The strict comparisons follow respectively from $32^4 < 5^4 2^{11}$ and $24^2 < 2 \cdot 17^2$.

Case $d \geq 3$. By (17), the coefficient per labeled xy -edge is at most

$$f(D, a) = 2^{-1/2-(d+D-a)/2}\Gamma^{d+D-2a}.$$

For fixed d , this decreases with D by ratio $3/4$ and increases with a by ratio $\sqrt{2}/\Gamma^2 > 1$. If x has at least two neighbors, $a \leq d - 1$ and $D \geq d$; summing over its d labeled incident edges gives

$$U + \sum_{y \sim x} T_y \leq 2^{-1/4}(3/4)^d + d 2^{-(d+2)/2}\Gamma^2. \quad (20)$$

If x has one neighbor, then $a = d$ and $D \geq d + 1$, since $v \geq 3$. The second term can then use Γ in place of Γ^2 , so (20) still applies. Both terms decrease for integer $d \geq 3$; for the second use $(d + 1)/(d\sqrt{2}) \leq 4/(3\sqrt{2}) < 1$. At $d = 3$ they satisfy

$$2^{-1/4}\frac{27}{64} < \frac{9}{25}, \quad 3 2^{-5/2}\Gamma^2 = \frac{27}{32\sqrt{2}} < \frac{3}{5}.$$

These follow from $75^4 < 2 \cdot 64^4$ and $135^2 < 2 \cdot 96^2$. Their sum is less than $24/25 < 1$.

Every connected graph on at least three vertices is therefore strict. Together with the small components and multiplicativity, this proves both the inequality and its equality characterization. $\square$

7.3 Shrinking hyperedges

We now transfer the graph estimate to hyperedges. Shrinking preserves every matching but can create the exceptional double-edge components of Lemma 7.2. The omitted incidences will pay for their correction factors, leaving a strict gain for every incidence beyond two per edge.

Definition 7.3 (Labeled multihypergraph matchings). A labeled multihypergraph F consists of n vertices and r labeled edges, each an ordinary subset of size at least two. Distinct labels may represent the same subset. Matchings are sets of labels whose subsets are pairwise disjoint. Components are defined by connectivity through hyperedges, with isolated vertices allowed. As for multigraphs, $Z(F)$ counts all matchings, including the empty one.

Lemma 7.4 (Incidence bound for hypergraph matchings). *Suppose F has no isolated component consisting of two identical size-two edges. Put*

$$\text{ex}(F) = \sum_{a \in E(F)} (|a| - 2), \quad B(F) = n + \sum_{a \in E(F)} |a| = n + 2r + \text{ex}(F).$$

Then

$$Z(F) \leq 2^{B(F)/4} \beta^{\text{ex}(F)}. \quad (21)$$

Proof. For each labeled edge a , choose any pair $p(a) \subseteq a$. Retain every label and form a multigraph G with these pairs on the same vertex set. An F -matching remains a G -matching. Its label set is unchanged, so distinct matchings remain distinct even when selected pairs coincide. Thus $Z(F) \leq Z(G)$.

There are exactly $\text{ex}(F)$ omitted incidences (a, w) , with $w \in a \setminus p(a)$. Such an incidence touches the component of G containing $p(a)$ and the component containing w , which may be the same. It therefore touches at most two double-edge components. Every double-edge component C of G is touched by an omitted incidence. Indeed, if none touched C , its two edge labels could not have omitted vertices, so both original hyperedges would be binary. No other hyperedge could intersect C : a selected pair outside C would make the intersection an omitted incidence, while a selected pair inside C would add a third edge. These two binary edges would form an isolated component of F , contrary to the assumption. Consequently

$$q(G) \leq 2 \text{ex}(F).$$

Applying Lemma 7.2, and using $\Gamma > 1$, yields

$$\begin{aligned} Z(F) &\leq 2^{(n+2r)/4} \Gamma^{q(G)} \leq 2^{(n+2r)/4} \Gamma^{2 \text{ex}(F)} \\ &= 2^{B(F)/4} \left(\frac{\Gamma^2}{2^{1/4}} \right)^{\text{ex}(F)}. \end{aligned}$$

Since $\Gamma^2/2^{1/4} = \beta$ and $\beta^4 = 6561/8192 < 1$, this proves the claim. $\square$

An isolated double edge has three matchings, exceeding $2^{6/4}$, so the excluded-component hypothesis is necessary. The argument permits overlapping omitted incidences: it uses only the bound of two exceptional components touched by each incidence.

7.4 Application to stable marriage

The hypergraph bound is independent of stable marriage. To apply it here, we need only connect antichains to matchings, identify the incidence budget, and rule out the forbidden isolated component. The last step is where monotone movement through stable partners enters. Following equality through the same steps will also give the classification.

Proof of Theorem 1.2. Form the labeled rotation hypergraph F of Definition 7.3 on M , with one edge for each rotation, consisting of its male participants. Two rotations sharing a man are comparable, so an antichain yields disjoint hyperedges. Together with the ideal–antichain bijection, this gives

$$|\text{Stab}(I)| = \#\{\text{antichains of } \mathcal{R}(I)\} \leq Z(F). \quad (22)$$

Precedence between rotations with disjoint participants can make this inequality strict.

In fact F has no isolated component on two men with two or more binary rotations. Otherwise take the first two of those rotations in a linear extension. The first swaps these men’s current partners. No intervening rotation can change either partner without involving that man, and the component assumption leaves no such intervening rotation. The second rotation swaps the same partners back, contradicting strict monotonicity along a man’s stable partners. External precedence constraints do not change this argument. In particular, Lemma 7.4 applies. Identity (14) identifies its $B(F)$ with m and its $\text{ex}(F)$ with $\text{ex}(I)$, proving (1).

For equality in $|\text{Stab}(I)| \leq 2^{m/4}$, we must have $\text{ex}(I) = 0$, since $\beta < 1$. The rotation hypergraph is then a multigraph. The previous paragraph excludes its isolated double and triple edges. Lemma 7.2 forces it to consist entirely of single-edge components covering all men; an isolated vertex would be strict. There are $r = n/2$ rotations and every subset is a hypergraph matching. Equality in (22) requires every subset to be an antichain, so the rotation poset itself is an antichain. Conversely, an antichain of binary rotations covering the men has 2^r ideals, and $m = n + 2r = 2n$, giving equality. Their female participant sets are also disjoint, since sharing a woman would imply comparability, and hence cover the women.

For sharpness, use t blocks of two men and two women. Within each block the orders are

$$\begin{aligned} u_0 : w_0 > w_1, & \quad u_1 : w_1 > w_0, \\ w_0 : u_1 > u_0, & \quad w_1 : u_0 > u_1. \end{aligned}$$

Place all internal partners before all external partners and complete the lists arbitrarily. If a perfect matching had an external pair, some block would have fewer than two internal pairs. That block would contain both a man and a woman matched externally, who would block with each other. Thus no stable matching uses external pairs. Both internal perfect matchings are stable, and any selection across blocks is stable. Therefore $n = 2t$, $m = 4t$, and $|\text{Stab}(I)| = 2^t$, ruling out every uniform base smaller than $2^{1/4}$. $\square$

Remark 7.5 (The stable-pair graph alone does not force equality). The antichain condition cannot be replaced merely by requiring the stable-pair graph to be a disjoint union of $K_{2,2}$ ’s. Here is a strict, complete four-by-four example, with labels 0, 1, 2, 3; each row is in decreasing preference order:

i	man i	woman i
0	0, 2, 1, 3	1, 0, 2, 3
1	1, 0, 2, 3	0, 1, 2, 3
2	2, 3, 0, 1	3, 0, 2, 1
3	3, 2, 0, 1	2, 3, 0, 1

Each of the two blocks supplies all four internal stable edges, but only the partner vectors $(0, 1, 2, 3)$, $(0, 1, 3, 2)$, and $(1, 0, 3, 2)$ are stable. The missing combination $(1, 0, 2, 3)$ is blocked by man 0 and woman 2. Thus $m = 8$ and $|\text{Stab}(I)| = 3 < 4$. Equality requires that every block combination be stable as well.

8 An application to Popular Roommates

The participant-count bound improves an existing enumeration algorithm for Popular Roommates. We use N for the total number of vertices of a roommates instance, and $c = 2078/625$. Preferences are strict over acceptable neighbors; every acceptable partner is preferred to being unmatched. A matching is popular if, against every other matching, the number of vertices preferring the other matching is at most the number preferring the current matching; indifferent vertices abstain.

Corollary 8.1. *Popular Roommates on N vertices can be solved in time $O^*(6.9717^N)$.*

This is a consequence of Theorem 1.1 and Kavitha’s algorithm [Kav25, Theorem 1 and Sections 3.3–3.4]. The counting theorem is stated for complete bipartite instances. We first extend its enumeration bound to incomplete lists, then account for the sizes of the two subproblems in Kavitha’s decomposition. For the incomplete-list step, we use the fact that all stable matchings in a strict incomplete bipartite instance match the same vertices. For completeness, suppose x is unmatched in μ but matched in ν . Follow the alternating component of $\mu \cup \nu$ starting at x . Stability of μ forces the next vertex to have a μ -partner whom it prefers to x . Stability of ν then forces that partner to have a preferred ν -partner. Repeating forces an endless alternating continuation, impossible in the finite path starting at a degree-one vertex. Interchanging μ, ν proves the assertion.

Find one stable matching by deferred acceptance; its matched vertices give the fixed matched sets. Let these sets have k vertices on each side. Restrict to them and append unacceptable partners below acceptable ones. Every old stable matching remains perfect and stable in this complete instance: an added edge is worse for both endpoints than their assigned acceptable partners. Restriction is injective. There are therefore fewer than c^k old matchings for $k \geq 1$, and one for $k = 0$. Moreover, complete-instance stable matchings can be enumerated in output-polynomial time [Gus87]; filtering for acceptable edges and stability in the original instance enumerates all old matchings within $O^*(c^k)$ time. Additional completions do not hurt this bound, since their total number is also bounded by $\text{SM}(k)$.

The two parts of Kavitha’s decomposition require different bipartite sizes. The algorithm ranges over subsets U with $|U| = i$. The stable-roommates enumeration on U uses a doubled bipartite instance with i vertices on each side. A roommates matching maps injectively to its two oriented copies. A blocking bipartite edge would give a blocking roommates edge; enumerating and filtering costs $O^*(c^i)$.

For the remaining $t = N - i$ vertices, Kavitha’s sign-witness enumeration [Kav25, Lemma 4] uses at most 2^t sign vectors. A nonempty class has $t/2$ vertices on each side of its induced bipartite instance. The stability-preserving injection and polynomial test in that lemma, together with the incomplete-list reduction above, give $O^*(2^t c^{t/2})$ time. Odd t contributes no perfect matching; retaining these terms in an upper bound is harmless. Witness classes need not be disjoint, since only an upper bound is used. The decomposition and polynomial popularity test thus give

$$O^*\left(\sum_{i=0}^N \binom{N}{i} c^i (2\sqrt{c})^{N-i}\right) = O^*((c + 2\sqrt{c})^N).$$

Finally, let $q = 69717/10000$. Since $q - c = 36469/10000 > 0$ and $(q - c)^2 - 4c = 67961/100000000 > 0$, we have $c + 2\sqrt{c} < q = 6.9717$, proving Corollary 8.1.

9 Conclusion

Revealed stable edges enclose conditional partner supports in ordered intervals that admit a geometric bound on their width. The resulting one-sided entropy estimate improves the participant-count rate, and rotation-boundary deletion recovers a further linear entropy loss. The stable-pair theorem uses the same participant-chain and no-return constraints to apply a general incidence bound for hypergraph matchings. In that bound, the incidences omitted when hyperedges are shrunk to pairs control the exceptional components created by the reduction.

The sharp stable-pair bound does not determine the participant-count growth constant: its sparse extremizers need not maximize the count at fixed n . The hypergraph relaxation discards precedence between disjoint rotations. Recovering these constraints while retaining their realization cost is one possible route to a stronger participant bound. A separate question is how close a near-extremal instance must be to an antichain of disjoint binary rotations covering the men.

A Hereditary fibers and marker thinning

We give the coordinatewise coupling used in the hereditary part of Theorem 1.4. The ordinary finite entropy chain is already proved in Lemma 4.1; the only additional issue is that edges fixed in advance produce deterministic markers rather than random-priority markers. Let $A \subseteq M$, put $U = M \setminus A$, and suppose that $\mathcal{S}_\eta \neq \emptyset$. Choose μ uniformly from $\mathcal{S}_\eta$ and reveal the coordinates in U in an independent random priority order. Applying the entropy chain to these coordinates gives

$$\log |\mathcal{S}_\eta| \leq \sum_{u \in U} \mathbb{E} \log X_u^\eta, \quad (23)$$

where X_u^η is the size of the partner support among stable matchings agreeing with η on A and with the sampled matching on the earlier coordinates of U .

Fix a sampled matching μ , a target $u \in U$, and a target priority $x \in (0, 1]$. Write the stable partners of u as $w_1 \succ_u \dots \succ_u w_q$, with $\mu(u) = w_j$, and for $h \neq j$ write $p_h = \mu^{-1}(w_h)$. The genuine marker at position h is

$$Z_h = \begin{cases} 1, & p_h \in A, \\ \mathbf{1}[T_{p_h} < x], & p_h \in U. \end{cases}$$

The matched men p_h are distinct and different from u . For every position with $p_h \in A$, introduce a fresh independent Bernoulli(x) bit D_h , and define

$$Y_h = \begin{cases} D_h, & p_h \in A, \\ \mathbf{1}[T_{p_h} < x], & p_h \in U. \end{cases}$$

Then $(Y_h)_{h \neq j}$ is iid Bernoulli(x), and $Y_h \leq Z_h$ coordinatewise. Passing from Z to Y only removes cuts. Its nearest left marker can therefore only move away from the center, and its nearest right marker can only do the same. The interval produced by Y contains the genuine revealed-pair interval.

Apply the common-prefix coupling of Section B to Y . At every fixed x ,

$$X_u^\eta \leq L + R - 1, \quad L, R \stackrel{\text{ind}}{\sim} \text{Geom}(x).$$

The calculation of Lemma 5.3 therefore gives $\mathbb{E} \log X_u^\eta \leq S_*$. Substitution into (23) proves $\log |\mathcal{S}_\eta| \leq |U|S_*$. If $U = \emptyset$, the prescribed male partner vector determines at most one matching. Nonemptiness gives $|\mathcal{S}_\eta| = 1 = C_*^0$, completing the boundary case.

B Marker windows and the common-prefix coupling

We prove Lemma 5.2 by realizing the finite marker lists as prefixes of infinite Bernoulli streams. The endpoint sentinels give the following finite waiting times. For a zero-one list $B = (B_1, \dots, B_m)$, define

$$\tau(B) = \min(\{r \in [m] : B_r = 1\} \cup \{m+1\}).$$

Thus $\tau(B) \geq 1$, even for an empty list. It equals the position of the first marker when one exists, and otherwise equals the sentinel position $m+1$.

Fix a stable-partner list $w_1, \dots, w_q$ with center j . The left list has length $j-1$ and scans indices $j-1, j-2, \dots, 1$. If its first marker is at distance r , then the nearest lower marker is $a = j-r$, so

$j - a = r = \tau(B^L)$. If it has no marker, $a = 0$ and $j - a = j = (j - 1) + 1 = \tau(B^L)$. The right list has length $q - j$, and the identical argument gives $b - j = \tau(B^R)$. Consequently,

$$b - a - 1 = \tau(B^L) + \tau(B^R) - 1.$$

This single calculation covers all four combinations of a present or absent marker on the two sides.

Appending extra entries never decreases τ : if the old list already has a marker, its first marker is unchanged; otherwise the first new marker has position at least $m + 1 = \tau(B)$. This lets us extend finite lists without decreasing their waiting times. Fix $x \in (0, 1]$. On $\Omega_x = \{0, 1\}^{\mathbb{N}} \times \{0, 1\}^{\mathbb{N}}$, take two independent iid Bernoulli(x) streams Y^L, Y^R . Their finite prefixes of lengths $j - 1$ and $q - j$ have exactly the same complete joint law as the genuine marker lists: the corresponding matched men are distinct, so their priority threshold events are independent Bernoulli(x) bits. We may therefore realize the genuine lists as those prefixes and regard the remaining coordinates as fresh dummy tails. Define the one-based first-success locations

$$L = \min\{r \geq 1 : Y_r^L = 1\}, \quad R = \min\{r \geq 1 : Y_r^R = 1\}.$$

For $x > 0$, they are finite almost surely, since the probability of m initial failures is $(1 - x)^m \rightarrow 0$. Their common mass function is $\mathbb{P}[L = r] = \mathbb{P}[R = r] = (1 - x)^{r-1}x$ for $r \geq 1$. They are independent because they depend on separate stream factors.

Let B^L be the finite prefix of Y^L . If that prefix has a success, its first success is also the first success of the infinite stream and $\tau(B^L) = L$. If the prefix has no success, then L lies beyond it, so $L \geq |B^L| + 1 = \tau(B^L)$. Hence $\tau(B^L) \leq L$ whenever L is finite. The same proof gives $\tau(B^R) \leq R$. Consequently,

$$|\mathcal{X}_u(\mu, A_u)| \leq b - a - 1 = \tau(B^L) + \tau(B^R) - 1 \leq L + R - 1.$$

The domination is therefore pointwise on a space where the finite lists and the infinite streams share their initial bits. At $x = 1$, both streams begin with a success almost surely and $L = R = 1$. At $x = 0$, the streams never succeed, so first-success variables are not used; that outer point is null in the target-priority integral.

C Exact comparisons for the stated bases

We prove both parts of Proposition 6.9 by bounding a finite sum of logarithms and its remaining tail. Positive series give rational bounds for the logarithms, and summation by parts gives a telescoping estimate for the tail.

C.1 Bounding the series by rational quantities

Put $s_k = 2 \log k / ((k + 1)(k + 2))$ for $k \geq 1$. Using $2 / ((k + 1)(k + 2)) = 2 / (k + 1) - 2 / (k + 2)$, finite summation by parts gives

$$\sum_{k=2}^K s_k = \frac{2}{3} \log 2 + \sum_{k=3}^K \frac{2}{k+1} \log \frac{k}{k-1} - \frac{2 \log K}{K+2}.$$

The boundary tends to zero. Relabelling $j = k - 3$ gives

$$S_* = \frac{2}{3} \log 2 + \sum_{j \geq 0} a_j, \quad a_j = \frac{2}{j+4} \log \left(1 + \frac{1}{j+2} \right). \quad (24)$$

All terms are nonnegative and are $O(j^{-2})$, so the limiting operations are justified by absolute convergence.

To bound the finite sum, we use the positive expansion, valid for $0 \leq z < 1$,

$$\log \frac{1+z}{1-z} = 2 \sum_{i \geq 0} \frac{z^{2i+1}}{2i+1}.$$

Define the rational function

$$A_{14}(z) = 2 \left(\sum_{i=0}^{13} \frac{z^{2i+1}}{2i+1} + \frac{z^{29}}{1-z^2} \right).$$

Every omitted denominator $2i+1$ is at least one; hence the positive remainder is at most the geometric tail and

$$\log \frac{1+z}{1-z} \leq A_{14}(z). \quad (25)$$

For an integer $m \geq 1$, set $z = (2m+1)^{-1}$. Then $(1+z)/(1-z) = 1 + 1/m$, so (25) is a rational upper bound for every logarithm in (24). The same choice with $m = 1$ bounds $\log 2$. For the lower direction, positivity of the omitted terms gives, for every $0 < z < 1$,

$$2 \sum_{i=0}^{15} \frac{z^{2i+1}}{2i+1} < \log \frac{1+z}{1-z}. \quad (26)$$

It remains to control the infinite tail. For $m \geq 1$, the Taylor bound

$$\log(1+t) \leq t - \frac{t^2}{2} + \frac{t^3}{3} \quad (t \geq 0)$$

follows by differentiating the difference between the two sides. Define $Q(m) = 2/m - 3/(2m^2) + 25/(18m^3)$. Substituting $t = 1/m$, multiplying by $2/(m+2)$, and clearing positive denominators gives

$$\frac{2}{m+2} \log \left(1 + \frac{1}{m} \right) \leq Q(m) - Q(m+1). \quad (27)$$

Indeed, after the cubic logarithm upper bound is inserted, the remaining slack equals a positive multiple of $90m^2 + 103m + 38$, divided by $18m^3(m+1)^3(m+2)$. The term a_j in (24) corresponds to $m = j+2$. Therefore the tail beginning at $j = 160$ satisfies

$$\sum_{j \geq 160} a_j \leq \sum_{m \geq 162} (Q(m) - Q(m+1)) = Q(162). \quad (28)$$

The last equality follows by finite telescoping and $Q(m) \rightarrow 0$.

C.2 Comparing the two target bases

Apply (25) to $\log 2$ and to $a_0, \dots, a_{159}$, and use (28). Substitution gives

$$\frac{2}{3} A_{14}(1/3) + \sum_{j=0}^{159} \frac{2}{j+4} A_{14} \left(\frac{1}{2(j+2)+1} \right) + Q(162) + 10^{-9} < \frac{120356492}{100000000}. \quad (29)$$

The inequality follows by clearing the positive denominators; the term 10^{-9} records a strict margin.

For $q = 1665987/500000$, take $z = (q - 1)/(q + 1) = 1165987/2165987$. Clearing denominators also gives

$$\frac{120356492}{100000000} + 10^{-9} < 2 \sum_{i=0}^{15} \frac{z^{2i+1}}{2i+1}. \quad (30)$$

Combining (24), (29), (30), and (26) yields

$$S_* < \frac{120356492}{100000000} < \log \frac{1+z}{1-z} = \log \frac{1665987}{500000}.$$

Since $C_* = e^{S_*}$ and $1665987/500000 = 3.331974$, this proves the first inequality in (13).

For the refined rate we reuse the same series upper bound, subtract a rational lower bound on the payment, and compare with the smaller target. The first four positive terms of (26) at $z = 1/3$ already give the rational inequality

$$\frac{693}{1000} < 2 \sum_{i=0}^3 \frac{(1/3)^{2i+1}}{2i+1} < \log 2. \quad (31)$$

Combining this with (29) and $\delta_{\text{ch}} = 616029 \log 2 / 196608000$ gives

$$\begin{aligned} S_* - \delta_{\text{ch}} &< \frac{120356492}{100000000} - \frac{616029}{196608000} \frac{693}{1000} \\ &= \frac{1968363197453}{1638400000000} < \frac{600697}{500000}. \end{aligned}$$

For the refined target, put $q = 2078/625$ and $z = (q - 1)/(q + 1) = 1453/2703$. Substitution into the sixteen-term lower estimate gives the finite rational certificate

$$\frac{600697}{500000} < 2 \sum_{i=0}^{15} \frac{z^{2i+1}}{2i+1} < \log \frac{1+z}{1-z} = \log \frac{2078}{625}. \quad (32)$$

The first inequality follows by clearing the positive denominator $500000 \prod_{i=0}^{15} (2i+1) 2703^{31}$; the remaining numerator is strictly positive. Together with (31) and $2078/625 = 3.3248$, this proves the second inequality in (13) and hence completes Proposition 6.9.

D Details for the global slack refinement

We give the chain-cut identity and logarithmic comparison used in Section 6. The rotation correspondence and monotone movement through stable partners are classical [IL86, GI89]; see Chapters 2–3 of [GI89].

D.1 Rotation orientation and chain cuts

The usual orientation records rotations eliminated from the man-optimal matching as an ideal. Reversing the poset and complementing each ideal gives the dual poset $\mathcal{R}$ used here. List the stable partners of u as $w_0(u), \dots, w_{q_u-1}(u)$, from best to worst. The rotations involving u form a chain of length $q_u - 1$. If the ideal D contains c members of this chain, then its matching μ_D assigns $w_{q_u-1-c}(u)$ to u . In particular, two ideals give u the same partner if and only if they have the same cut on the u -chain.

Lemma D.1 (Chain-cut forcing). *Suppose $\rho \in D$ is a rotation involving a man p . If an ideal E satisfies $\mu_E(p) = \mu_D(p)$, then $\rho \in E$. More generally, if fixing p 's partner forces a rotation $\sigma \in E$ with $\rho < \sigma$, then ideal closure forces $\rho \in E$.*

Proof. An ideal meets the rotations involving p in an initial segment. Equality of the assigned partners is equality of the two chain cuts, so every rotation of the p -chain contained in D , including ρ , is also contained in E . The second assertion is the downset property. $\square$

If $\mu_D(u) = w_j(u)$, then $\rho(D, u)$ in the main text is the last included canonical rotation on the u -chain. Its preferred endpoint for u is $w_j(u)$, and its other endpoint is $w_{j+1}(u)$. Thus keeping $\rho(D, u)$ in an ideal prevents the partner index of u from moving past j , exactly as used in Lemma 6.3.

D.2 Exact tail comparison

We compare the binary choice exponent in the deletion bound with the binomial entropy term at the threshold used in Lemma 6.8. Let $0 < \alpha < 1/2$ and $k \leq \alpha n$. The binomial theorem gives the standard estimate

$$\sum_{i=0}^k \binom{n}{i} \leq e^{nh(\alpha)}. \quad (33)$$

Indeed, for every $i \leq k$, $\alpha^i(1-\alpha)^{n-i} \geq \alpha^{\alpha n}(1-\alpha)^{(1-\alpha)n} = e^{-nh(\alpha)}$; multiply the sum by this common lower bound and compare with $(\alpha + 1 - \alpha)^n = 1$.

For $\alpha = 47/1250$, exact rational logarithm bounds give

$$h(\alpha) < \frac{11559}{50000} \log 2. \quad (34)$$

Since $1/4 - \alpha/2 = 11560/50000$, we obtain

$$\left(\frac{1}{4} - \frac{\alpha}{2}\right) \log 2 - h(\alpha) > \frac{\log 2}{50000}. \quad (35)$$

To verify (34), use

$$\begin{aligned} h(\alpha) &= \frac{47}{1250} \log \frac{1250}{47} + \frac{1203}{1250} \log \frac{1250}{1203}, \\ \log \frac{1250}{47} &= 4 \log 2 + 2 \log \frac{5}{4} + \log \frac{50}{47}, \quad \log \frac{1250}{1203} = \log \frac{1 + 47/2453}{1 - 47/2453}. \end{aligned}$$

Apply the positive atanh upper bound (25) to $1/3, 1/9, 3/97, 47/2453$. If the resulting rational upper bound is denoted by H^+ , direct cross-multiplication gives

$$H^+ < \frac{11559}{50000} \frac{6931471}{10000000}.$$

The first seven positive atanh terms at $1/3$ give the independent rational lower bound

$$\frac{6931471}{10000000} < 2 \sum_{i=0}^6 \frac{(1/3)^{2i+1}}{2i+1} < \log 2.$$

These two rational comparisons prove (34) and hence the entropy margin used in Lemma 6.8.

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